A carry-counting approach to the $k = 3$ case of Erdős Problem 727

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Abstract

We present a candidate unconditional proof that infinitely many positive integers n satisfy $((n+3)!)^2 \mid (2n)!$, and hence also $((n+2)!)^2 \mid (2n)!$. The quadratic $f(t) = 210t^2 + 391t + 179$ makes each of $f(t)+1, f(t)+2, f(t)+3$ a product of two linear forms. A fixed arithmetic progression handles small primes; direct carry counting, uniform quadratic exponential-sum estimates, and a finite residue calculation bound the remaining failures. The only external analytic input is the classical prime number theorem in fixed arithmetic progressions. The argument gives a positive proportion of successful parameters on the progression and $\gg \sqrt{Z}$ successful integers $n \le Z$.

Status. This is a candidate proof circulated for scrutiny. The accompanying Lean development is complete on the author's machine (see the closing section); independent expert verification and independent reproduction of that build remain pending. The general case of arbitrary fixed k is not claimed here.

1 Statement and strategy

Erdős, Graham, Ruzsa and Straus asked whether, for every fixed $k \ge 2$, there are infinitely many n for which $((n+k)!)^2 \mid (2n)!$ [1, p. 90]. We consider the case $k = 3$, using

$$f(t) = 210t^2 + 391t + 179.$$

Theorem 1.1. *There exist fixed integers $Q \ge 1$, $0 \le t_0 < Q$, and $X_0 \ge 1$ such that for every integer $X \ge X_0$, at least $X/100$ integers $v \in [X, 2X)$ satisfy*

$$\big((f(t_0 + Qv) + 3)!\big)^2 \mid \big(2f(t_0 + Qv)\big)!. \tag{1.1}$$

Consequently there is a constant $c_* > 0$ such that, for all sufficiently large Z,

$$\#\{1 \le n \le Z : ((n+3)!)^2 \mid (2n)!\} \ge c_*\sqrt{Z}.$$

Corollary 1.2. *There are infinitely many positive integers n such that $((n+2)!)^2 \mid (2n)!$.*

Write $A(n) = (n+1)(n+2)(n+3)$ and $B(n) = \binom{2n}{n}$. Since $(n+3)! = n!A(n)$ and $(2n)! = (n!)^2 B(n)$, cancellation gives

$$((n+3)!)^2 \mid (2n)! \iff A(n)^2 \mid B(n). \tag{1.2}$$

For a prime p, Legendre's formula yields the carry identity

$$c_p(n) := v_p(B(n)) = \sum_{h \ge 1} \left(\left\lfloor \frac{2n}{p^h} \right\rfloor - 2 \left\lfloor \frac{n}{p^h} \right\rfloor \right) = \sum_{h \ge 1} \mathbf{1}_{\{n/p^h\} \ge 1/2}. \tag{1.3}$$

Thus it suffices to prove $c_p(n) \ge 2v_p(A(n))$ for every prime dividing $A(n)$. We arrange this for all bounded primes by congruences, then bound the union of the remaining failure events. No independence between primes or between the six linear factors is assumed.

Here $\{x\} = x - \lfloor x \rfloor$, $e(x) = \exp(2\pi i x)$, and $\|x\|$ denotes distance to the nearest integer. All sums indexed by p are over primes. All moduli, constants, and exponent margins are fixed before $X \to \infty$; implicit constants may depend on them.

2 The polynomial family and a fixed progression

The three factorizations are

$$\begin{aligned} f(t) + 1 &= (35t + 36)(6t + 5), \\ f(t) + 2 &= (t + 1)(210t + 181), \\ f(t) + 3 &= (15t + 14)(14t + 13). \end{aligned} \tag{2.1}$$

Let $L_i(t) = a_i t + b_i$, $1 \le i \le 6$, denote the displayed factors, and let $j_i \in \{1, 2, 3\}$ be the associated shift. Table 1 records all constants needed below. Direct expansion verifies

$$f(t) = \alpha_i L_i(t)^2 + \beta_i L_i(t) - j_i, \qquad f'(-b_i/a_i) = c_i. \tag{2.2}$$

Table 1: The six factors; d_i is the reduced denominator of α_i.

i	$L_i(t)$	j_i	c_i	α_i	β_i	d_i
1	$35t + 36$	1	-41	$6/35$	$-41/35$	35
2	$6t + 5$	1	41	$35/6$	$41/6$	6
3	$t + 1$	2	-29	210	-29	1
4	$210t + 181$	2	29	$1/210$	$29/210$	210
5	$15t + 14$	3	-1	$14/15$	$-1/15$	15
6	$14t + 13$	3	1	$15/14$	$1/14$	14

Every L_i is primitive, and all prime factors of the slopes lie in $\{2, 3, 5, 7\}$. The within-pair resultants have absolute values $41, 29, 1$. A prime dividing factors from different pairs divides a nonzero difference between two of $f + 1, f + 2, f + 3$, so is at most 2. Consequently a prime $p > 1000$ divides at most one $L_i(t)$, and its root class satisfies $f'(t) \equiv c_i \not\equiv 0 \pmod{p}$.

2.1 Small primes by lifting and the Chinese remainder theorem

Set $Y = 1000$ and $G(t) = \prod_i L_i(t) = A(f(t))$. Since $G(0) = 180 \cdot 181 \cdot 182$, its prime support is $S = \{2, 3, 5, 7, 13, 181\}$. For every prime $p \le Y$ outside S, impose $t \equiv 0 \pmod{p}$, ensuring $p \nmid G(t)$. For $p \in S$, set $e_p = v_p(G(0))$ and choose a lifting depth λ_p as follows:

p	2	3	5	7	13	181
e_p	3	2	1	1	1	1
λ_p	3	3	2	2	2	2
$\lambda_p + 2e_p$	9	7	4	4	4	4

Each λ_p is larger than every individual valuation $v_p(179 + j)$, $1 \le j \le 3$. If r_p is the least nonnegative residue of 179 modulo p^{λ_p}, prescribe

$$f(t) \equiv r_p + p^{\lambda_p}(p^{2e_p} - 1) \pmod{p^{\lambda_p + 2e_p}}, \qquad t \equiv 0 \pmod{p}. \tag{2.3}$$

This has a unique lift of the root $t = 0 \pmod{p}$, because $f'(0) = 391 = 17 \cdot 23$ is coprime to every $p \in S$. Indeed, for $r \ge 1$,

$$f(u + zp^r) \equiv f(u) + zp^r f'(u) \pmod{p^{r+1}};$$

when the derivative is a unit, varying $z \pmod{p}$ realizes each next target digit exactly once.

Condition (2.3) preserves each $v_p(f(t) + j) = v_p(179 + j)$ and prescribes $2e_p$ consecutive digits equal to $p - 1$ in positions $\lambda_p, \dots, \lambda_p + 2e_p - 1$ of $f(t)$. Each produces a carry on doubling, even for $p = 2$, regardless of the incoming carry. Hence $c_p(f(t)) \geq 2e_p = 2v_p(G(t))$. The Chinese remainder theorem now gives $0 \leq t_0 < Q$, with

$$Q = \prod_{p \in S} p^{\lambda_p + 2e_p} \prod_{\substack{p \leq Y \\ p \notin S}} p, \tag{2.4}$$

such that every $t = t_0 + Qv$ satisfies the required inequalities for all $p \leq Y$. The modulus is fixed, however large it may be.

Write

$$F(v) = f(t_0 + Qv), \qquad \mathcal{L}_i(v) = L_i(t_0 + Qv).$$

For $v \in [X, 2X)$, all six forms are positive and at most CX, for some fixed $C \geq 1$, and $F(v) \asymp X^2$. For $p > Y$, the slope $a_i Q$ is a unit modulo every power of p.

2.2 Repeated factors and failure events

Discard parameters for which $p^2 \mid \mathcal{L}_i(v)$ for some i and $p > Y$. Each condition is one class modulo p^2, so their union has size

$$E_{\mathrm{sq}}(X) \leq 6X \sum_{p > Y} \frac{1}{p^2} + 6\pi(\sqrt{CX}) \leq \frac{6X}{Y} + o(X) = \frac{3}{500} X + o(X). \tag{2.5}$$

Here $\sum_{m > Y} m^{-2} \leq 1/Y$, and $\pi(z) \leq z$ suffices for the endpoint error. For a remaining parameter, every prime $p > Y$ dividing $G(t)$ has exponent exactly one. If $p \mid \mathcal{L}_i(v)$, then $F(v) \equiv -j_i \pmod{p}$; since $p > 6$, this supplies a carry at level 1. It remains to exclude

$$p \mid \mathcal{L}_i(v), \qquad \{F(v)/p^h\} < \tfrac{1}{2} \quad \text{for every } h \geq 2. \tag{2.6}$$

We bound the union of these events in the ranges below.

3 Prime harmonic sums

The only external analytic theorem used here is the classical estimate

$$\pi(x; d, a) = \frac{\mathrm{Li}(x)}{\varphi(d)} + O_d\big(x \exp(-c_d \sqrt{\log x})\big), \qquad (a, d) = 1, \tag{3.1}$$

for fixed d, with $c_d > 0$; see [2, §27.12, (27.12.5) and (27.12.8)]. Here $\pi(x; d, a)$ counts primes in the indicated class, φ is Euler's totient, and $\mathrm{Li}(x) = \int_2^x du/\log u$. In particular, $\pi(CX) = o(X)$ for fixed C.

Lemma 3.1. *For fixed d, a with $(a, d) = 1$,*

$$\sum_{\substack{p \leq z \\ p \equiv a \ (d)}} \frac{1}{p} = \frac{\log \log z}{\varphi(d)} + B_{d,a} + o(1). \tag{3.2}$$

Consequently, if R is a fixed set of reduced classes modulo d, $0 < \alpha < \beta$, and $C > 0$ is fixed, then

$$\sum_{\substack{X^\alpha < p \leq CX^\beta \\ p \bmod d \in R}} \frac{1}{p} = \frac{|R|}{\varphi(d)} \log \frac{\beta}{\alpha} + o(1). \tag{3.3}$$

Also, for fixed $P > 1$, $c > 0$, and $\delta > 0$,

$$\lim_{X \to \infty} \sum_{P < p \leq X^\delta} \frac{1}{p} \exp\left(-\frac{c \log X}{\log p}\right) = \int_0^\delta e^{-c/y} \frac{dy}{y} = E_1(c/\delta), \tag{3.4}$$

where $E_1(z) = \int_z^\infty e^{-w}\, dw/w$.

Proof. Partial summation writes the left side of (3.2) as $\pi(z; d, a)/z + \int_2^z \pi(u; d, a)u^{-2}\,du$. The main term in (3.1) gives $(\log\log z)/\varphi(d)$ plus a constant; the error integral converges because $\int_2^\infty e^{-c_d\sqrt{\log u}}\,du/u < \infty$. Taking differences gives (3.3), including the unrestricted prime sum when $d = 1$.

For (3.4), write the unrestricted sum as $H(z) = \log\log z + B + R_0(z)$, where $R_0(z) \to 0$, and integrate $g_X(z) = \exp(-c\log X/\log z)$ against $dH(z)$ on $(P, X^\delta]$. The main term is $\int_{\log P/\log X}^{\delta} e^{-c/y}\,dy/y$; its missing lower tail tends to zero. To control the error, fix $Z > P$. The part below Z tends to zero as $X \to \infty$. On $[Z, X^\delta]$, integration by parts and monotonicity of g_X bound the error by $2\sup_{z\geq Z}|R_0(z)|g_X(X^\delta)$. First let $X \to \infty$, then $Z \to \infty$. The substitution $w = c/y$ gives the last equality in (3.4). □

4 Very small primes

Fix

$$\delta = \frac{1}{20}, \qquad \rho = \frac{1001}{2000}, \qquad c = \frac{1}{2}\log(1/\rho).$$

For $Y < p \leq X^\delta$, put $\ell = \lfloor \log X/(2\log p)\rfloor$. Then $\ell \geq 10$ and $p^\ell \leq \sqrt{X}$. On the unique root class of $\mathcal{L}_i \pmod p$, $F'(v) \equiv Qc_i \not\equiv 0 \pmod p$. The lifting argument above therefore makes $v \mapsto F(v)$ a bijection from that class modulo p^ℓ onto the residues congruent to $-j_i \pmod p$.

The units digit is $p - j_i$ and already gives a carry. To avoid a carry at the next level, there are $(p-1)/2$ choices for the next digit; with no incoming carry thereafter, each later digit has $(p+1)/2$ choices. Thus the number of residues avoiding every carry at levels $2, \ldots, \ell$ is

$$\frac{p-1}{2}\left(\frac{p+1}{2}\right)^{\ell-2} \leq p^{\ell-1}\rho^{\ell-1}.$$

Each residue occurs at most $X/p^\ell + 1$ times, giving

$$\#\{v \in [X, 2X) : \text{(2.6) holds for } i, p\} \leq \left(\frac{X}{p} + p^{\ell-1}\right)\rho^{\ell-1}. \tag{4.1}$$

The sum of the second terms, over all primes in this range and all six forms, is $O(\sqrt{X}\log\log X) = o(X)$. Since

$$\ell - 1 \geq \frac{\log X}{2\log p} - 2, \qquad \rho^{\ell-1} \leq \rho^{-2}\exp\left(-\frac{c\log X}{\log p}\right),$$

Lemma 3.1 bounds the union count by

$$\limsup_{X\to\infty}\frac{E_{\mathrm{small}}(X)}{X} \leq 6\rho^{-2}E_1(c/\delta) < \frac{1}{100}. \tag{4.2}$$

For the strict numerical bound, $\rho^{-2} < 4$, $c/\delta = 10\log(2000/1001) > 6$, and $E_1(z) \leq e^{-z}/z$; hence the expression is less than $4e^{-6} < 1/100$. The comparisons follow, without numerical integration, from $\log x \geq 2(x-1)/(x+1)$ for $x \geq 1$ and $e^3 > \sum_{r=0}^{8} 3^r/r! > 20$.

5 Medium primes and uniformity

5.1 A quadratic exponential-sum bound

Lemma 5.1. *Let $q \geq 3$ be odd, $(a, q) = 1$, and let I be an interval of L consecutive integers. Uniformly in these data and in $\theta \in \mathbb{R}$,*

$$\left|\sum_{r\in I} e\left(\frac{ar^2}{q} + \theta r\right)\right| \ll \left(\frac{L}{\sqrt{q}} + \sqrt{q}\right)\log(2q). \tag{5.1}$$

Proof. Suppose first $L \le q$, and translate I to $0 \le r < L$. For $b \pmod q$, set $G_b = \sum_{s \bmod q} e((as^2 - bs)/q)$. Squaring and putting the difference of the two variables equal to h gives

$$|G_b|^2 = \sum_{h \bmod q} e\left(\frac{ah^2 - bh}{q}\right) \sum_{s \bmod q} e\left(\frac{2ahs}{q}\right) = q,$$

since $2a$ is a unit modulo q. Fourier inversion yields

$$\sum_{r=0}^{L-1} e\left(\frac{ar^2}{q} + \theta r\right) = \frac{1}{q} \sum_{b \bmod q} G_b \sum_{r=0}^{L-1} e\big((\theta + b/q)r\big).$$

The inner absolute value is at most $\min(L, (2\|\theta + b/q\|)^{-1})$, interpreted as L at integral frequencies. The points $\theta + b/q \pmod 1$ are $1/q$-spaced, so only a bounded number lie in each distance shell of width $1/q$ from an integer. Summing the geometric-sum bounds gives $O(q \log(2q))$, uniformly in θ. Since $|G_b| = \sqrt{q}$, the desired short-interval bound is $O(\sqrt{q} \log(2q))$. For general L, divide I into at most $L/q + 1$ intervals of length at most q. □

5.2 Uniform distribution in each band

Fix $\eta = 10^{-6}$ and $\gamma_j = 2/j$, $4 \le j \le 40$. For $J = 4, \dots, 39$, use the band

$$X^{\gamma_{J+1}+\eta} < p \le X^{\gamma_J - \eta}. \tag{5.2}$$

These exponent intervals are nonempty, since $\gamma_J - \gamma_{J+1} \ge 1/780 > 2\eta$.

Fix i and choose the least nonnegative root v_0 of $\mathcal{L}_i \pmod p$. Conditional on $p \mid \mathcal{L}_i(v)$, write $v = v_0 + pz$, where z runs over $L = X/p + O(1)$ consecutive integers. With $D = 210Q^2$,

$$F(v_0 + pz) = Dp^2z^2 + pF'(v_0)z + F(v_0). \tag{5.3}$$

We show that the vectors

$$\big(\{F(v)/p^2\}, \dots, \{F(v)/p^J\}\big)$$

are asymptotically uniformly distributed in $[0,1)^{J-1}$, *uniformly over primes in the band.*

Fix a nonzero integer frequency $(h_2, \dots, h_J)$, and let H be its highest nonzero index. If $H \ge 3$, the quadratic coefficient of the phase $\sum_{r=2}^{J} h_r F(v_0 + pz)/p^r$ has reduced denominator $q = p^{H-2}$, for all sufficiently large X. Indeed, its numerator modulo p is $Dh_H \not\equiv 0 \pmod p$ once p exceeds the prime factors of the fixed nonzero integer Dh_H. Lemma 5.1, allowing the arbitrary real linear coefficient, gives the normalized bound

$$O\left(\left(p^{-(H-2)/2} + \frac{p^{H/2}}{X}\right)\log X\right) = o(1) \tag{5.4}$$

uniformly in (5.2): here $p \ge X^{\delta}$, $H \le J$, and $p^{H/2}/X \le p^{J/2}/X \le X^{-\eta J/2}$. If $H = 2$, the quadratic coefficient is the integer Dh_2, and the nonconstant phase reduces to $h_2F'(v_0)z/p$. Because $F'(v_0) \equiv Qc_i \pmod p$, its numerator modulo p is the fixed nonzero integer h_2Qc_i. The geometric sum is $O(p)$, with normalized bound

$$O(p/L) = O(p^2/X) = o(1), \tag{5.5}$$

uniformly, since $p \le X^{1/2-\eta}$.

For completeness, fixed-frequency cancellation suffices uniformly for the box $[0,1/2)^{J-1}$. Approximate its indicator above and below by continuous periodic functions whose integrals differ arbitrarily little, then approximate those functions by finite trigonometric polynomials. At each stage only finitely many frequencies occur, so (5.4)–(5.5) hold uniformly for all of them. First let $X \to \infty$, then refine the approximation. It follows that, uniformly over the band, the proportion of conditioned parameters without a carry at levels $2, \dots, J$ is

$$2^{1-J} + o(1). \tag{5.6}$$

Summing (5.6) times $X/p + O(1)$ over the primes incurs only $o(X)$: the reciprocal-prime sum in a fixed band is bounded by Lemma 3.1, the discrepancy is uniform, and the endpoint terms total $O(\pi(X^{1/2-\eta})) = o(X)$. Summing the six forms and all 36 bands therefore gives

$$\begin{aligned}\limsup_{X\to\infty} \frac{E_{\mathrm{med}}(X)}{X} &\le 6 \sum_{J=4}^{39} 2^{1-J} \log \frac{J+1}{J} \\ &\le 6 \cdot \frac{1}{4} \sum_{J=4}^{\infty} 2^{1-J} = \frac{3}{8}.\end{aligned} \tag{5.7}$$

We used $\log(1 + 1/J) \le 1/J \le 1/4$.

5.3 Boundary strips

The prime exponents between $\delta = 1/20$ and $1/2 + \eta$ not covered by (5.2) lie in the 37 strips

$$X^{\gamma_j-\eta} \le p \le X^{\gamma_j+\eta}, \qquad 4 \le j \le 40.$$

Count every event $p \mid \mathcal{L}_i(v)$ in these strips, ignoring carries. Lemma 3.1 and $X/p + O(1)$ give

$$\begin{aligned}\limsup_{X\to\infty} \frac{E_{\mathrm{bdry}}(X)}{X} &\le 6 \sum_{j=4}^{40} \log \frac{\gamma_j + \eta}{\gamma_j - \eta} \\ &\le 12\eta \sum_{j=4}^{40} \frac{1}{\gamma_j - \eta} \le \frac{6\eta \cdot 814}{1 - 20\eta} < \frac{1}{100}.\end{aligned} \tag{5.8}$$

Here $\sum_{j=4}^{40} j = 814$; endpoint errors are $o(X)$, since all primes involved are at most $X^{1/2+\eta}$. Any overlaps at boundaries are harmless in these union bounds.

6 Large primes and a finite residue calculation

Suppose $p > X^{1/2+\eta}$ divides $L_i(t)$, with $t = t_0 + Qv$, and write $L_i(t) = pm$. Then $m \ge 1$, and uniformly in this range

$$\frac{m}{p} \le \frac{CX}{p^2} = O(X^{-2\eta}) = o(1), \qquad \frac{f(t)}{p^2} = \alpha_i m^2 + \beta_i \frac{m}{p} - \frac{j_i}{p^2}. \tag{6.1}$$

For $i \ne 3$, Table 1 shows that $d_i \mid a_i$ and $(b_i, d_i) = 1$. Reducing $L_i(t) = pm$ modulo d_i gives

$$m \equiv b_i p^{-1} \pmod{d_i}. \tag{6.2}$$

As the unit class of p varies, this permutes the unit classes of m. The relation uses the original coefficients a_i, b_i, so restricting t to the CRT progression does not change it.

The possible fractional parts of $\alpha_i m^2$, for unit m, are listed in Table 2. Each distinct value has equally many preimages: squaring is a homomorphism on the finite unit group with constant fiber size, and the numerator of α_i is a unit modulo d_i. Thus the last column is the fraction of unit classes for which this fractional part is below $1/2$.

No listed fraction is 0 or $1/2$. The finite sets have positive distance from both the integers and $1/2$. By (6.1), the correction tends uniformly to zero, so for sufficiently large X it neither crosses $1/2$ nor wraps around an integer. Every listed fraction above $1/2$ therefore supplies a carry at level p^2. Failures lie in a fixed set R_i of reduced prime classes modulo d_i, with $|R_i|/\varphi(d_i) = \theta_i$, by (6.2).

The form $i = 3$ is favorable for a different reason:

$$\frac{f(t)}{p^2} = 210m^2 - \frac{29m}{p} - \frac{2}{p^2}.$$

Table 2: Large-prime residues. Divide the listed numerators by d_i to obtain the fractional parts.

i	d_i	Numerators of the fractional parts	θ_i
1	35	$6, 19, 24, 26, 31, 34$	$1/6$
2	6	5	0
4	210	$1, 79, 109, 121, 151, 169$	$1/3$
5	15	$11, 14$	0
6	14	$1, 9, 11$	$1/3$

Its correction is strictly negative and tends uniformly to zero. It lies in $(-1/2, 0)$ for large X, so the fractional part lies in $(1/2, 1)$. This always gives the second carry; put $\theta_3 = 0$.

Since $p \le CX$ whenever $p \mid \mathcal{L}_i(v)$, the large-prime union count satisfies

$$E_{\text{large}}(X) \le \sum_{i \ne 3} \sum_{\substack{X^{1/2+\eta} < p \le CX \\ p \bmod d_i \in R_i}} \left(\frac{X}{p} + O(1) \right).$$

The total endpoint error is $O(\pi(CX)) = o(X)$, even though C may be large, because it is fixed. Lemma 3.1 and $\sum_i \theta_i = 1/6 + 1/3 + 1/3 = 5/6$ give

$$\limsup_{X\to\infty} \frac{E_{\text{large}}(X)}{X} \le \frac{5}{6} \log \frac{1}{1/2+\eta} < \frac{5}{6} \log 2 < \frac{7}{12}. \tag{6.3}$$

The last inequality follows from $\log 2 < 7/10$; the first four terms of the exponential series already give $e^{7/10} > 2$.

7 Completion of the proof

Proof of Theorem 1.1 and Corollary 1.2. All primes $p \le Y$ satisfy the carry inequalities throughout the fixed progression. After the repeated-factor exclusions, every failure at a larger prime lies in (2.6). The very small range, medium bands, boundary strips, and large range cover every such prime, and no prime divisor of any $\mathcal{L}_i(v)$ exceeds CX.

Let $E(X)$ be the number of unsuccessful parameters in $[X, 2X)$. The union bound, together with (2.5), (4.2), (5.7), (5.8), and (6.3), gives

$$\limsup_{X\to\infty} \frac{E(X)}{X} < \frac{3}{500} + \frac{1}{100} + \frac{3}{8} + \frac{1}{100} + \frac{7}{12} = \frac{2953}{3000} < \frac{99}{100}. \tag{7.1}$$

All constants and dimensions are fixed; the Fourier approximations were fixed before taking each asymptotic limit and only then refined. The finitely many estimates therefore hold simultaneously on a common tail. For every sufficiently large integer X, at least $X/100$ of the X integers in $[X, 2X)$ succeed. By (1.2) this is exactly (1.1).

The map $v \mapsto F(v)$ is strictly increasing for positive v, so these parameters give infinitely many distinct successful integers. Also $F(v) \le C_2 v^2$ for some fixed $C_2 > 0$ and all sufficiently large v. Given large Z, take

$$X = \left\lfloor \frac{1}{2} \sqrt{Z/C_2} \right\rfloor.$$

Every $v \in [X, 2X)$ then has $F(v) \le Z$, and at least $X/100 \gg \sqrt{Z}$ of these distinct values succeed. This proves the counting assertion. Finally, $((n+2)!)^2 \mid ((n+3)!)^2$, so every successful n for $k = 3$ is also successful for $k = 2$. □

Relation to the accompanying Lean development

This note condenses the original candidate manuscript of 6 September 2026, retaining its analytic route and its five-term failure budget. The accompanying Lean source follows a reorganized proof: explicit local witnesses replace iterated lifting, proved Mertens estimates in fixed progressions replace the stronger quoted prime number theorem, and finite Fourier analysis replaces the limiting box approximation. Its failure budget also differs: the boundary strips are absorbed into the medium range, and the four range bounds are slackened to $0.01 + 0.02 + 0.35 + 0.60 = 0.98 < 1$, which suffices for infinitude. It should not be described as a line-by-line certification of this presentation.

The source endpoints `Erdos727.erdos727_k3` and `Erdos727.erdos727_k2` express the intended infinitude statements. The former unfolds to

```
Set.Infinite {n : ℕ |
  (Nat.factorial (n + 3))^2 | Nat.factorial (2 * n)}
```

The stronger quantitative conclusions above are not separately exported as named theorems in that development. At commit `c0100a2` of the repository, every module compiles without `sorry`, and `lake build` reports that `Erdos727.erdos727_k2` depends only on the axioms `propext`, `Classical.choice`, and `Quot.sound`; a build-time assertion enforces the same for `erdos727_k3`. This was checked on the author's machine only; independent reproduction of the build has not been recorded, and the argument presented here does not depend on it. Independent expert review remains pending.

AI assistance and provenance

This work was developed with assistance from `gpt-6-astra`, `fable 5.1`, and `gemini-3.8`. Responsibility for the mathematical claims and the final manuscript rests with the author.

References

[1] P. Erdős, R. L. Graham, I. Z. Ruzsa, and E. G. Straus, *On the prime factors of* $\binom{2n}{n}$, Mathematics of Computation **29** (1975), no. 129, 83–92. The factorial-divisibility question appears on p. 90. Original article.

[2] NIST Digital Library of Mathematical Functions, *§27.12: Asymptotic Formulas: Primes*, equations (27.12.5) and (27.12.8). `https://dlmf.nist.gov/27.12`.